香港教育大學
The Education University of Hong Kong

Module Title：	INT2065 Foundations of Information and Communication Technologies
Medium of Instruction:	English
Time:	1 Sep 2026 – 13 Dec 2026, 18:30 – 21:20 (Tue)
Venue:	D3-LP-03
Lecturers:	Dr. Davy Tsz Kit Ng (davyngtk@eduhk.hk)

1. Course Synopsis

This course provides students with fundamental knowledge and skills in information and communication technologies (ICTs) so that they are competent to study more advanced ICT-related subjects.

2. Course Intended Learning Outcomes (CILOs)

Upon completion of this course, students will be able to:

CILO1 state and explain different hardware and software concepts of a computer system, and data representation inside computer;

CILO2 explain the concepts of modern networks and Internet;

CILO3 acquire the knowledge and skills for different applications software; and

CILO4 examine the impact of ICT in society.

3. Content, CILOs, and Teaching & Learning Activities

Course Content	CILOs	Suggested Teaching & Learning Activities
Introduction to Computer Hardware and Software: • Understanding digital components; • The language of computers: bits and bytes; • Input devices, output devices, processing, storage, and connectivity; • Operating system fundamentals; • System software and application software; • Collaborative application software. • Using AI to facilitate the teaching/learning to deliver the subject knowledge regarding computer hardware and software.	$CILO_1$	Lectures, workshop, hands-on practices
Principles of Data Communication and Network: • Data communication fundamentals; • Networking fundamentals; • Network architectures and components; • Client/server network and topologies; • The Internet and its applications; • Digital devices and media. • Using AI to facilitate the teaching/learning to deliver the subject knowledge regarding principles of data communication and network.	$CILO_{1,3}$	Lectures, hands-on practices
Introduction to Software Programming: • Understanding the importance of software programming; • The life cycle of software development; • Exploring programming languages; • Basic HTML, JavaScript, and Cascading Style Sheets programming. • Using AI to facilitate the teaching/learning to deliver the subject knowledge regarding software programming.	$CILO_{2,3}$	Lectures, hands-on practices
Impact of IT and AI in Society: • Computer viruses; • Cybercrime and identity theft;	$CILO_4$	Lectures, case study analysis, on-line discussion

• Cryptography; • Privacy and security: • copyright, and ethical issue; • Impact of information and communication technology; • Technology and innovation for the future of products. • Considering the critical and ethical concerns behind when engaging with AI to learn and teach these computer concepts, guided by Education Bureau (EDB) (2026)'s AI Literacy Learning Framework for Primary and Secondary Schools.		

Course schedule

Session	Class Date	Session Overview
Part I: Lecture on concepts and theories		
1	1 Sep 18:30-21:20	• Introduction • Recent Digital Education Policies in Hong Kong
2	8 Sep 18:30-21:20	• Hardware, HTML • HKEAA Documents
3	15 Sep 18:30-21:20	• Software • Lesson Plan Design
4	22 Sep 18:30-21:20	• Database • Worksheet Design • Sharing from Oracle University
5	29 Sep 18:30-21:20	• Internet • Sharing from Local ICT Teacher
	2 Oct (Fri) 09:30-12:20	• Microbit and Huskeylens making workshop (NEW ACADEMIC BUILDING N-2/F-16) (Quota: 15)
6	6 Oct 18:30-21:20	• Network • IT Infrastructure and Network in Schools Ying Wa College, No. 1 Ying Wa Street, Shamshuipo, Kowloon
7	13 Oct 18:30-21:20	• CL and ICL Curriculum Refinement and Development Ying Wa College, No. 1 Ying Wa Street, Shamshuipo, Kowloon
8	20 Oct 18:30-21:20	• Software development • Project Consultation
9	27 Oct 18:30-21:20	• Invited Sharing from publisher • Online micro-teaching simulation
10*	3 Nov	• Self-paced reading and online videos
Project consultation, presentation and debriefing		
	10 Nov 16:00-18:00	EDB Sharing at Kowloon Tong (Optional) (Quota: 10-15)
11	17 Nov 18:30-21:20	• Group Presentation
12	24 Nov 18:30-21:20	• Group Presentation • Course wrap-up
13	28 Nov (Sat) 15:30-17:30	Database Training - Room201 at EdUHK (TKO Center) (Optional) (Quota: 10-15)

*Students are not required to attend face-to-face lectures.

Presentation materials for Assignment 1 must be submitted on or before 10 Nov 2026 23:59.

Assignment 2 must be completed and uploaded to Moodle by 8 Dec 2026 23:59.

Group and Individual Assignments

Assessment Tasks	Weighting (%)	CILO
a. Participation: Student attendance in classroom activities, online learning activities and collaborative activities.	10%	$CILO_{1,2,3,4}$
b. Group project: Propose a solution to teach an ICT knowledge. The solution can be an AI application. A 10-minute presentation (plus 5-minute Q&A) of the work is required, which should cover the relevant computer concept(s), objectives, pedagogical ground, pre/in/post-class instructional design, a tool to teach the computer concept, a tool and its educational value behind, as well as its ethical and social implications (if any). Each group should consist of 4-6 students. Students need to submit (1) a lesson plan, (2) worksheet(s), (3) (AI-augmented) resources (e.g., vibe coding app to visualize a computer concept, gamified apps) and digital tools. Please submit group member list on or before 29 Sep 2026, 23:59. A Google Form link will be provided.	40%	$CILO_{1,2,3,4}$
c. Individual Project (Mini Research Evaluation): Building on their group project outcomes, students will conduct and report (in 1,500 English words) a small-scale formative evaluation by interviewing a relevant stakeholder (e.g., a teacher, student/classmates) about the course design, feasibility, proposed learning outcomes, and its ethical and social implications (if any); students will synthesize interview insights to refine their design, and reflect on how the (AI and) digital tool may support or hinder meaningful learner development.	50%	$CILO_{1,2,3,4}$

Use of Generative AI in Course Assessments

In this course, generative AI tools may be used in some or all assessment tasks. Instructors will provide specific instructions, including any restrictions or additional requirements (e.g., proper acknowledgment, reflective reports), during the first lesson and in relevant assessment briefs.

Last update:
1 Sep 2026

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General Grade Descriptor

Generic Grade Descriptors

Course Grade	A+	A	A-	B+	B	B-	C+	C	C-	D	F
Attainment of Course Intended Learning Outcomes (CILOs) / assessment criteria	All / most of the stated CILOs /assessment criteria have been met at an exceptionally high level	All / most of the stated CILOs / assessment criteria have been fully met at a high standard		All / most of the stated CILOs / assessment criteria have been fully met at a good standard			Most of the stated CILOs / assessment criteria have been met		Some of the stated CILOs / assessment criteria have been met		None of the stated CILO / assessment criteria have been met
Interpretation	**DISTINCTION**			**GOOD**			**SATISFACTORY**		**BELOW SATISFACTORY**		**FAIL**
	Exceptional	**Outstanding**	**Excellent**	**Very Good**	**Good**	**Fairly Good**	**Satisfactory**	**Barely Satisfactory**	**Below Satisfactory**	**Barely Pass**	**Fail**
Grade Point Equivalent	4.33	4.00	3.67	3.33	3.00	2.67	2.33	2.00	1.67	1.00	0.00
Mark Range	96-100	88-95	81-87	76-80	71-75	66-70	59-65	51-58	46-50	41-45	0-40
Understanding of subject matter/ topic	Demonstrates a deep understanding and full interpretation of the subject matter/topic and the underlying theories.	Demonstrates a thorough understanding and a highly convincing interpretation of the subject matter/topic and the underlying theories.	Demonstrates a full understanding and interpretation of the subject matter/topics and the underlying theories.	Demonstrates a high level of understanding and interpretation of the subject matter/topics with a strong grasp of the underlying theories.	Demonstrates a good level of understanding and interpretation of the subject matter/topics and underlying theories.	Demonstrates a general understanding and interpretation of the subject matter/topics and the underlying theories.	Demonstrates a fairly adequate grasp of the subject matter/topics in terms of factual understanding but theoretical understanding is at a superficial level.	Demonstrates a basic understanding of the subject matter/ topic but with a superficial grasp of theoretical perspectives.	Demonstrates an inadequate grasp of the subject matter/topic and with little theoretical discussion.	Demonstrates very inadequate understanding of the subject matter/topic with little theoretical discussion.	Demonstrates poor understanding of the subject matter/topics and the underlying theories.
Cognitive/ Intellectual skills	Shows a very sophisticated level of critical reflection, analysis, evaluation and/or synthesis.	Shows a very high level of critical reflection, analysis, evaluation and/or synthesis.	Shows a high level of critical reflection, analysis, evaluation and/or synthesis.	Shows a good level of critical reflection, analysis, evaluation and/or synthesis.	Shows a sound level of critical reflection, analysis, evaluation and/or synthesis.	Shows some trace of critical reflection, analysis, evaluation and/or synthesis.	Shows limited level of critical reflection, with little or no attempt at analysis, evaluation and/or synthesis, though the material is organized logically.	Shows limited level of critical reflection, with little or no attempt at analysis, evaluation and/or synthesis, though the material is organized logically.	Shows very little evidence of critical reflection, with no attempt at analysis, evaluation and/or synthesis, though the material is organized logically.	Shows no evidence of critical reflection and no attempt at analysis, evaluation and/or synthesis.	Shows no evidence of critical reflection and no attempt at analysis, evaluation, and/or synthesis.
Presentation of ideas	Presents highly original ideas with great lucidity and succinctness.	Presents original ideas with great lucidity and succinctness.	Presents fairly original ideas with great lucidity and succinctness.	Presents very sound ideas with lucidity and succinctness.	Presents sound ideas with reasonable clarity.	Presents sound ideas with reasonable clarity.	Presents some ideas with limited soundness and clarity.	Presents some ideas with limited soundness and clarity.	Presents ideas with very limited clarity.	Presents ideas with ambiguity.	Presents ideas poorly and ambiguously.
Organization and referencing	Extremely well organized and structured, fluently and accurately written and correctly referenced.	Very well organized and structured, fluently and accurately written and correctly referenced.	Very well organized and structured, fluently written and correctly referenced.	Well organized, fluently written and, in the main, correctly referenced.	Well organized, fluently written and, in the main, correctly referenced.	Fairly well organized, fluently written and, in the main, correctly referenced.	Recognizable organization, and referenced with errors.	Loosely organized, weak grammatically and referenced with errors.	Poorly organized, very weak grammatically and referenced with frequent errors.	Poorly organized, very weak grammatically and referenced with frequent errors.	Poorly organized, full of grammatical errors, communication is seriously impeded, and reference with numerous errors.
Use of readings/ literature	Evidence of extensive and judicious use of relevant readings/ literature to support the arguments.	Evidence of extensive and thoughtful reading/ literature in the subject/topic area.	Evidence of a substantial knowledge of relevant readings/ literature.	Evidence of extensive readings/ literature in the topic area.	Evidence of adequate readings/ literature in the topic area.	Evidence of some readings/ literature.	Evidence of some readings/ literature.	Evidence of a somewhat cursory acquaintance with readings/ literature.	Evidence of a minimal acquaintance with readings/ literature.	Very minimal acquaintance with readings/ literature.	No evidence of independent reading.